

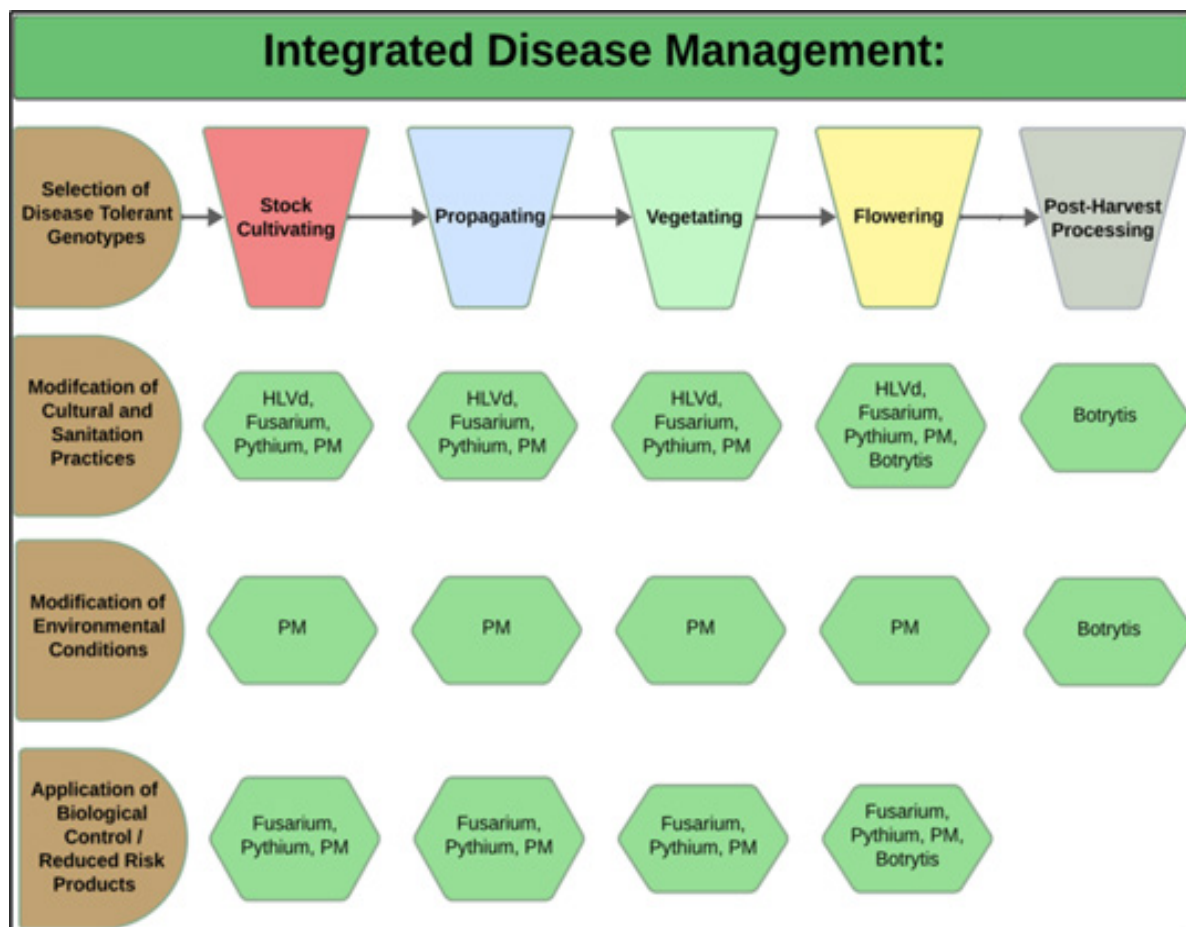


Figure 3. Integrated disease management strategies (left panel, in brown) are developed according to the crop development stage (top panel). The hexagons (in green) illustrate the specific diseases being targeted, which are discussed in more detail below. HLVd = hop latent viroid; PM = powdery mildew; *Botrytis cinerea* = bud rot.

We also discuss aspects of the microbial colonization of cannabis inflorescences by yeasts and molds and propose IDM strategies to reduce the total microflora present. Monitoring the microbial colonization of inflorescences is an important quality aspect of cannabis, which is under strict regulatory control and presents a unique and challenging component of crop management that is not found in most other crops [19,20]. This review should aid in the design or refinement of further IDM programs in greenhouse-cultivated cannabis operations. Detailed descriptions of the symptoms caused by various pathogens at different stages of cannabis growth during commercial production and the approaches that can be taken to manage them are described below.

2. Cannabis Pathogens: Symptoms and Management Approaches at Different Stages of Growth

2.1. Stock Cultivation Stage

Stock (mother) plants provide a source of vegetative cuttings, which are commonly used in commercial cannabis production. These plants generally constitute a range of genotypes that are chosen for their desired phenotypic characteristics and biochemical profiles. They are grown in designated areas within the greenhouse or in separate indoor rooms. Physical separation of stock plants from larger-scale commercial production is important to prevent the spread of pathogens. The ages of these stock plants can vary and typically range from 3 to 12 months, depending on the facility. In the context of disease development, older plants often exhibit signs of declining growth, such as reduced shoot